

	perpendicular to the magnetic field, and the current flowing in the wire, which are all kept the same as before.
5(a)	Gravitational potential ϕ at a point is the work done per unit mass in bringing a small test mass from infinity to that point.
5(b)	To escape from the planet's surface, an object needs to have sufficient kinetic energy in order to overcome the gravitational pull of the planet of mass M. The minimum kinetic energy required is equivalent to the gain in gravitational potential energy by the object of mass m in leaving planet to escape to infinity, not to return to the planet. loss in KE = gain in GPE $\frac{1}{2}mv^2 - 0 = U_{\infty} - U_{\text{planet's surface}} \quad (\text{assuming that KE at infinity is zero})$ $\frac{1}{2}mv^2 = 0 - \left(-\frac{GMm}{R} \right) \quad (\text{since } U_{\infty} = 0)$ $v^2 = \frac{2GM}{R}$ $v = \sqrt{\frac{2GM}{R}}$
5(c)(i) 5(c)(ii)	

Qn	Solution
6(a)(i)	The mean KE of a gas molecule is directly proportional to the thermodynamic